

VeloCity — Reference Context Map

Trainer-only material. This is the reference solution for the VeloCity context mapping exercise. Do not hand it to the domain expert or the participants — they work only from `velocity-vision.md`.

The solution below is one well-argued answer, not the only correct one. See **Accepted alternatives** before judging a group's result.

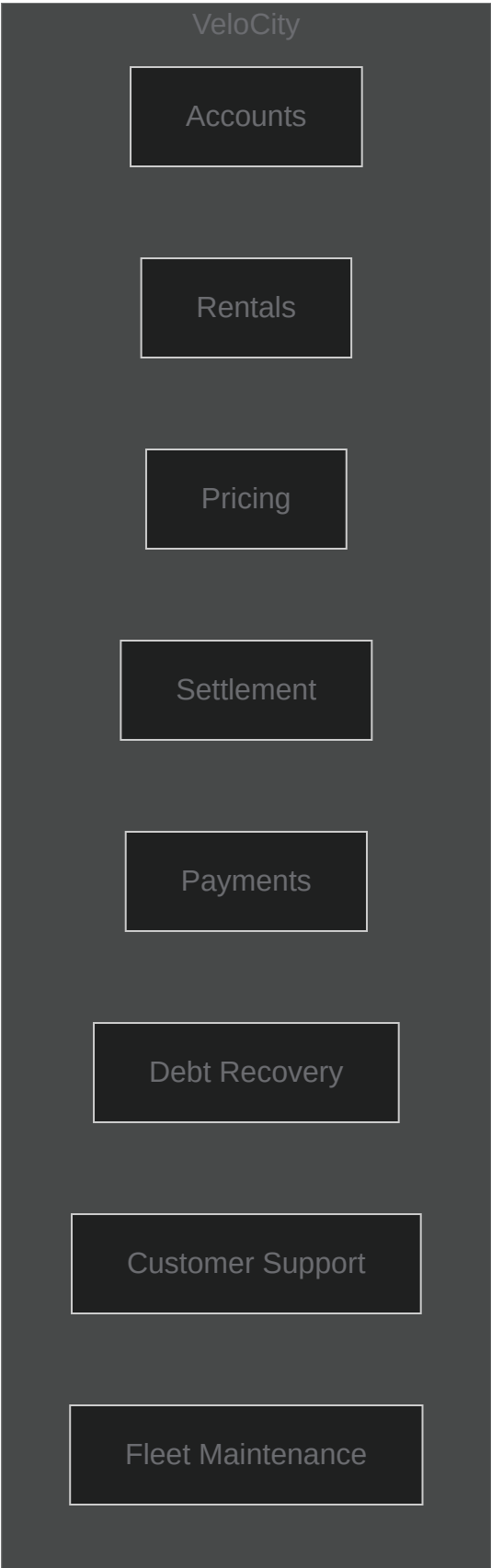
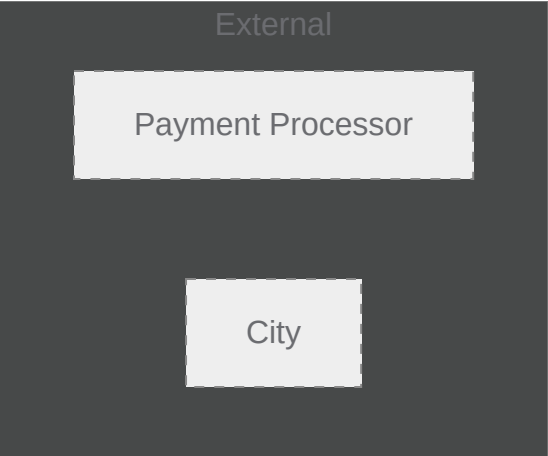
Bounded Contexts

Context	Owns	Key language
Accounts	Onboarding, account activation, account blocks and unblocks	user, registration, activated, blocked
Rentals	Bike availability (available / withdrawn), reserve & unlock for any authorised client, rental sessions, station returns & locking, max-rental-time end, withdrawing a bike on a fault it detects	reservation, unlock, rental session, available, withdrawn
Pricing	Pure calculation of the total fee — tiered hourly fees, penalty, drop zone / unauthorized zone fees, tariff valid at session start	free period, hourly fee, penalty, drop zone fee, total fee
Settlement	Charging each finished ride, retries, rider's balance and outstanding debt	charge, settled, outstanding debt
Payments	Payment method verification and charging; shields VeloCity from the external processor	payment method, pre-authorization, verified, charge
Debt Recovery	Active pursuit of outstanding debt — reminders, payment plans, pause / resume	recovery process, paused, resumed, exhausted, payment plan

Context	Owns	Key language
Customer Support	Incidents: reported → type detected → assigned → accepted / rejected → resolved	incident, accept, reject, escalate, reopen, resolve
Fleet Maintenance	Broken bikes, maintenance, deciding when a bike returns to circulation	broken, maintenance, fixed, back in circulation

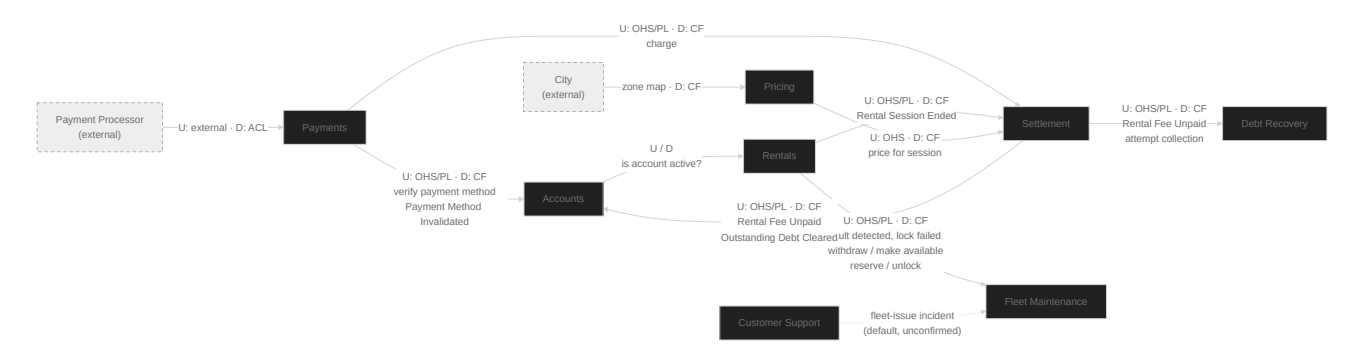
External parties: the **Payment Processor** and the **City** (zone map used by Pricing).

Overview



Context Map

Every arrow points from **upstream (U)** to **downstream (D)**. Labels give the pattern and what flows across the boundary. Data may flow in both directions (events one way, commands the other) — the arrow shows **whose language the integration uses**.

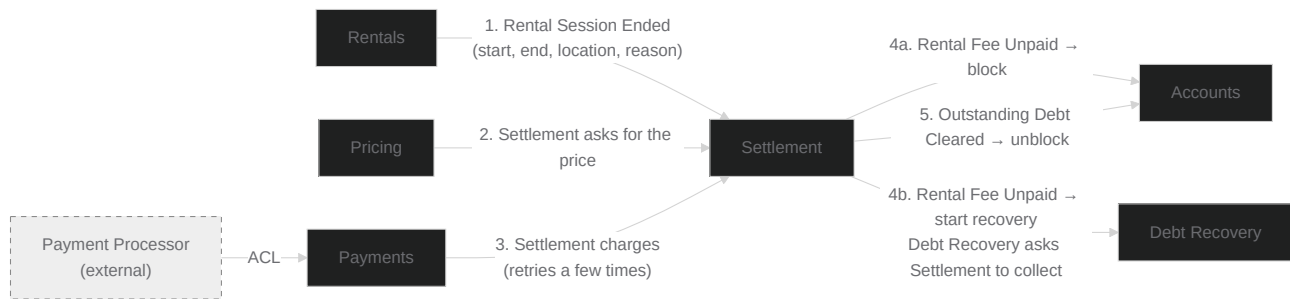


Legend

Abbreviation	Meaning
U / D	Upstream / Downstream
OHS	Open Host Service — upstream offers a protocol for any number of clients
PL	Published Language — the shared, documented format of that protocol (events, commands)
CF	Conformist — downstream adopts the upstream model as-is
ACL	Anticorruption Layer — downstream translates the upstream model into its own
dashed arrow	Default assumption, not confirmed

Zoom: from a finished ride to money

The ride-to-money flow has the most traffic in both directions, yet no cycles. Settlement *calls* Pricing and Payments, and Debt Recovery *calls* Settlement — the caller is always downstream.



Key Design Decisions

- **No Station context.** A station is a thin client of Rentals; locking, fault detection at return and locking failures live in Rentals.
- **No reroute.** Every return ends the ride; the location only decides the fee (drop zone fee vs. unauthorized zone return fee). The client-reported location is trusted.
- **Rentals owns availability in generic language** — *available / withdrawn*, never *broken*. Fleet Maintenance (and potentially other clients, e.g. a future marketing context re-branding bikes) decides what happens to a bike and calls Rentals.
- **Rentals withdraws a bike itself when it detects a fault** (locking failure, fault at station return). Only clients decide when it becomes available again. A broken bike is never offered to riders.
- **Reserve / unlock is a generic Rentals capability.** Only a rider's unlock starts a rental session; Fleet Maintenance reacts to a technician's unlock with *Maintenance Started*.
- **A locking failure ends the session immediately** with an explanation; the rider is not charged extra and the bike is withdrawn.
- **Max rental time is detected by Rentals**, which ends the session automatically; Pricing adds the penalty.
- **Pricing is a pure calculator**, called by Settlement, using the tariff valid at session start.
- **Settlement is the only context that moves money** (via Payments) and owns the rider's balance. Debt Recovery asks Settlement to collect.
- **Account blocks have exactly two reasons, each from one source:** invalid card (Payments) and outstanding debt (Settlement). Settlement drives both block and unblock for debt.
- **No cycles.** If two contexts have arrows in both directions, ask whose language the integration uses — if both, it is either a Partnership or a misplaced boundary.

Accepted Alternatives for the Debrief

- A single **Returns** context (station + zone returns) instead of returns inside Rentals.

- **"Billing"** as the name for Pricing — it is the domain expert's word.
- **Pricing reacting** to *Rental Session Ended* instead of being called by Settlement (orchestration direction is a design choice, not a heuristic outcome).
- **Fleet Maintenance upstream of Rentals** (Rentals reports faults via Maintenance's OHS) — as long as there is no cycle.
- The **Payment Processor drawn as external** with ACLs directly in Accounts and Settlement, without a Payments context.
- **Debt Recovery** → **Accounts** for unblocking (discuss: *whose fact is "the debt is paid"?*).
- **Settlement and Debt Recovery merged**, if the group can't find a language difference — point them to *reminders, payment plans, pause vs. charge, retry, settled*.

Deferred / Open

- **Billing disputes** → **pausing Debt Recovery** (from Customer Support) — not modelled yet.
- **Fleet-issue incidents** routed Customer Support → Fleet Maintenance → Rentals (withdraw) — default, unconfirmed.
- **Faults signalled by the bike itself** received by Rentals (it owns locks and stations) — default, unconfirmed.

Material Follow-ups

velocity-vision.md is already updated. Still to do:

- ☐ Re-export velocity-vision.pdf .
- ☐ Board (velocity-processes.jpg):
 - ☐ Remove Bike Return To Drop Zone Rejected → Bike Rerouted ; add Rental Session Ended .
 - ☐ Replace Bike Locking Failed → Customer Support Contacted with Bike Locking Failed → Rental Session Ended + Bike Fault Reported .
 - ☐ Remove the Account Blocked Due To Multiple Failed Payments lane.
 - ☐ Add Rental Fee Charged / Rental Fee Charge Failed .
 - ☐ Unlock failed → reservation *released* (not expired).
 - ☐ Fix the dangling Incident Rejected so every branch ends in *Incident Resolved*.